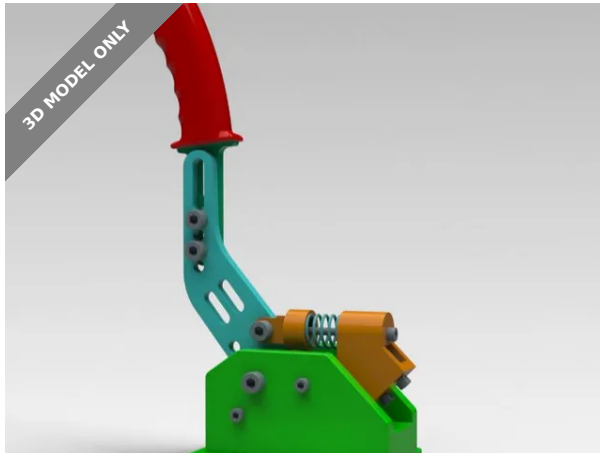




Sim Racing Handbrake - Load Cell

 eruiec

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Summary

Sim racing handbrake with load cell.

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This project is a remix of the mentioned potentiometer-based sim racing handbrake, upgraded to use a load cell for force-based measurement instead of position-based input.

The result is a much more realistic handbrake feel, similar to real hydraulic handbrakes used in motorsport.

- Compatible with Arduino Uno using UnoJoy
- Adjustable stiffness depending on spring and preload
- Uses mostly 3D-printed parts

Hardware used:

- Load cell (typically 10–50 kg (<https://www.sparkfun.com/load-cell-10kg-straight-bar-tal220.html#content-features>))
- HX711 load cell amplifier
- Springs (user preference)
- 2xM4x18 bolts

- 6xM5x18 bolts + 4xM5 nut
- 3xM5x70 bolts + 3xM5 nut
- 3xM8x30 bolts + 3xM8 nut
- M8x70 bolt + M8 nut
- Arduino with USB cable

Firmware:

- Arduino Uno flashed with UnoJoy (see unojoy instructions: <https://github.com/AlanChatham/UnoJoy>)

Calibration:

- Windows Game Controller settings
- In-game control settings
- Tested with Assetto Corsa

Disclaimer:

This is a DIY project. Assembly and use are at your own risk. Ensure proper wiring and secure mounting before use.

This remix is based on



Sim Racing Handbrake V2
by Thornyx

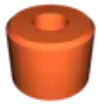
Model files



2025-11-30-spacer.stl



2025-12-28-upper-handle.stl



2025-12-28-spacer-2.stl



2025-11-30-slider-joint.stl



2025-11-30-joint-spring.stl



2025-11-30-clamp.stl



2025-11-25-clamp-screw.stl



2025-12-28-lower-handle.stl



2025-12-11-handle.stl



2025-12-11-handle-2.stl



2025-12-28-base.stl

Other files



2025-12-28-handbrake-loadcell.ino

📄 Arduino and unojoy code

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